

Arachnophobia

Arachnophobia is a specific phobia brought about by the irrational fear of spiders and other arachnids such as scorpions.^[3]

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Signs and symptoms

People with arachnophobia tend to feel uneasy in any area they believe could harbour spiders or that has visible signs of their presence, such as webs. If an arachnophobe sees a spider, they may not enter the general vicinity until they have overcome the panic attack that is often associated with their phobia. Some people scream, cry, have emotional outbursts, experience trouble breathing, sweat and experience increased heart rates when they come in contact with an area near spiders or their webs. In some extreme cases, even a picture, a toy, or a realistic drawing of a spider can trigger intense fear.

Reasons

Arachnophobia may be an exaggerated form of an instinctive response that helped early humans to survive^[4] or a cultural phenomenon that is most common in predominantly European societies.^[5]

Evolutionary

Arachnophobia

Other names

Arachnephobia^[1]



Though many arachnids are harmless, a person with arachnophobia may still panic or feel uneasy around one. Sometimes, even an object resembling a spider can trigger a panic attack in an arachnophobic individual. The above cartoon is a depiction of the nursery rhyme "Little Miss Muffet", in which the title character is "frightened away" by a spider.

Pronunciation

/əˌræknəˈfoʊbiə/

Specialty

Psychiatry

Treatment

Exposure therapy^[2]

An evolutionary reason for the phobia remains unresolved. One view, especially held in evolutionary psychology, is that the presence of venomous spiders led to the evolution of a fear of spiders, or made acquisition of a fear of spiders especially easy. Like all traits, there is variability in the intensity of fear of spiders, and those with more intense fears are classified as phobic. Being relatively small, spiders do not fit the usual criterion for a threat in the animal kingdom where size is a factor, but they can have medically significant venom and/or cause skin irritation with their setae.^[6] However, a phobia is an irrational fear as opposed to a rational fear.^[3]

By ensuring that their surroundings were free from spiders, arachnophobes would have had a reduced risk of being bitten in ancestral environments, giving them a slight advantage over non-arachnophobes in terms of survival. However, having a disproportionate fear of spiders in comparison to other, potentially dangerous creatures^[7] present during *Homo sapiens'* environment of evolutionary adaptiveness may have had drawbacks.

In *The Handbook of the Emotions* (1993), psychologist Arne Öhman studied pairing an unconditioned stimulus with evolutionarily-relevant fear-response neutral stimuli (snakes and spiders) versus evolutionarily-irrelevant fear-response neutral stimuli (mushrooms, flowers, physical representation of polyhedra, firearms, and electrical outlets) on human subjects and found that ophidiophobia and arachnophobia required only one pairing to develop a conditioned response while mycophobia, anthophobia, phobias of physical representations of polyhedra, firearms, and electrical outlets required multiple pairings and went extinct without continued conditioning while the conditioned ophidiophobia and arachnophobia were permanent.^[8]

Psychiatrist Randolph M. Nesse notes that while conditioned fear responses to evolutionarily novel dangerous objects such as electrical outlets is possible, the conditioning is slower because such cues have no prewired connection to fear, noting further that despite the emphasis of the risks of speeding and drunk driving in driver's education, it alone does not provide reliable protection against traffic collisions and that nearly one-quarter of all deaths in 2014 of people aged 15 to 24 in the United States were in traffic collisions.^[9] Nesse, psychiatrist Isaac Marks, and evolutionary biologist George C. Williams have noted that people with systematically deficient responses to various adaptive phobias (e.g. arachnophobia, ophidiophobia, basophobia) are more temperamentally careless and more likely to end up in potentially fatal accidents and have proposed that such deficient phobia should be classified as "**hypophobia**" due to its selfish genetic consequences.^{[10][11][12][13]}

A 2001 study found that people could detect images of spiders among images of flowers and mushrooms more quickly than they could detect images of flowers or mushrooms among images of spiders. The researchers suggested that this was because fast response to spiders was more relevant to human evolution.^[14]

Cultural

An alternative view is that the dangers, such as from spiders, are overrated and not sufficient to influence evolution. Instead, inheriting phobias would have restrictive and debilitating effects upon survival, rather than being an aid. For some communities such as in Papua New Guinea and Cambodia spiders are included in traditional foods. This suggests arachnophobia may, at least in part, be a cultural, rather than genetic trait.^{[15][16]}

Treatments

The fear of spiders can be treated by any of the general techniques suggested for specific phobias. The first line of treatment is systematic desensitization – also known as exposure therapy.^[2] Before engaging in systematic desensitization, it is common to train the individual with arachnophobia in relaxation techniques, which will help keep the patient calm. Systematic desensitization can be done in vivo (with live spiders) or by getting the individual to imagine situations involving spiders, then modelling interaction with spiders for the person affected and eventually interacting with real spiders. This technique can be effective in just one session, although it generally takes more time.^[17]

Recent advances in technology have enabled the use of virtual or augmented reality spiders for use in therapy. These techniques have proven to be effective.^[18] It has been suggested that exposure to short clips from the Spider-Man movies may help to reduce an individual's arachnophobia.^[19]

Epidemiology

Arachnophobia affects 3.5 to 6.1 percent of the global population.^[20]

See also

- Apiphobia (fear of bees)
- Entomophobia (fear of insects)
- Myrmecophobia (fear of ants)

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This page was last edited on 31 March 2022, at 20:52 (UTC).

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